# RAVINTHRA A

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## PROFESSIONAL SUMMARY

MCA fresher with hands-on experience building and deploying production-grade Python applications. Skilled in Django, FastAPI, REST API design, PostgreSQL, and applied Machine Learning (Scikit-learn, BERT/PyTorch). Built 4 end-to-end deployed projects spanning fintech risk modeling, NLP, and full-stack web development — demonstrating strong ability to design, train, evaluate, and serve ML models with real-world deployment on Docker and Render. Seeking a backend or ML engineering role to contribute scalable solutions from day one.

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript

**Backend:** Django, FastAPI, REST API Design

**Databases:** PostgreSQL (ORM, Query Optimization)

**Machine Learning:** Scikit-learn, XGBoost, LightGBM, BERT (PyTorch), SHAP, Feature Engineering

**Tools:** Docker, Git, Gunicorn, Render, WhiteNoise, Hugging Face Spaces

**Core Concepts:** OOP, API Architecture, Microservices, Business-aware Threshold Optimization

## PROJECTS

### Alumni Information System
Python | Django 5.0 | PostgreSQL | Gunicorn | WhiteNoise | Render

- Full-stack Django web app with 4 custom roles (Super Admin, Content Manager, Event Manager, Alumni) featuring admin-approval registration, OTP-based password recovery, and email authentication backend.
- Built alumni directory with search/filter by batch, major, and location; inline profile editing with social links and profile pictures; bulk alumni import via Excel/CSV upload.
- Deployed on Render using Gunicorn + WhiteNoise via render.yaml blueprint; production-ready settings with environment-variable-based configuration and collectstatic build script.

### Customer Churn Prediction System
Python | Scikit-learn | Django REST | Random Forest | Render

- Trained and compared 4 ML models (Logistic Regression, Decision Tree, Random Forest, SVM) on Telco dataset; Random Forest selected with ROC-AUC 0.8576 and 81.34% accuracy; model + scaler serialized for production.
- Built Django REST API with health-check endpoint, CORS, rotating log files, request-ID tracing, and Pydantic-style serializer validation; covered by 9 unit and integration tests.
- Live web UI at customer-churn-prediction-ezz6.onrender.com serving real-time predictions with confidence scores and top-3 feature importance explainability.

### Credit Risk / Loan Default Prediction
Python | Django | FastAPI | XGBoost | LightGBM | SHAP | Docker | Render

- SQL-first feature engineering from 4 normalized tables (customers, loans, credit_history, payments) generating 66 features including DTI ratio, credit utilization, payment delays, and MNAR-aware missingness indicators.
- Compared 4 models (Logistic Regression, XGBoost, LightGBM, GBM); Logistic Regression selected as champion with CV AUC 0.8494 for regulatory interpretability; resolved feature leakage reducing AUC from 0.999 to realistic 0.83.
- Business-aware threshold (0.11) via asymmetric cost model (FN=Rs.1L, FP=Rs.10K) yielding Rs.91.2L savings; Django web app + FastAPI microservice via Docker Compose with SHAP explainability; live at credit-risk-ai.onrender.com.

### AI Resume Analyzer
Python | BERT | PyTorch | Django | Docker | Hugging Face Spaces

- Fine-tuned bert-base-uncased (110M params) with Linear(768→5) + Dropout(0.3) classification head using AdamW optimizer and linear warmup scheduler on 600+ synthetic resumes; achieved validation F1 of 1.0 across 5 tech roles.
- Core pipeline computes BERT [CLS] embedding cosine similarity between resume and job description for semantic match scoring; skill gap analysis via regex keyword extraction and set operations (intersection and difference).
- Security-hardened Django backend with CSRF protection, XSS/Clickjacking headers, magic-byte PDF validation, and 5MB upload limit; Dockerized for Hugging Face Spaces (CPU PyTorch) and Render via render.yaml.

## EDUCATION

**Master of Computer Applications (MCA)** — **CGPA: 8.49**
Hindusthan College of Engineering and Technology — 2023 – 2025

**Bachelor of Science in Computer Science (B.Sc CS)** — **CGPA: 7.93**
Gobi Arts and Science College — 2020 – 2023